

LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

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Problem Statement #24 — Smart Cities, Transport & Logistics Public

Bus Live Tracking & Crowding Estimator

1. Problem Context

The proposed system is a Public Bus Live Tracking & Crowding Estimator designed for a municipal transit environment. The system collects GPS data from city buses to track their current locations and estimate their arrival times at upcoming stops. It also processes ticketing sensor data to estimate the passenger crowding level of buses.

The platform provides useful real-time information to commuters and operational monitoring capabilities to Fleet Controllers. The system must process data from a large number of active buses efficiently and provide reliable and secure transit information.

2. System Purpose

The main purpose of the system is to:

- Track the real-time location of city buses using GPS data.
- Calculate estimated arrival times (ETA) for upcoming bus stops.
- Estimate passenger crowding levels using ticketing sensor data.
- Provide commuters with live bus location, ETA, and crowding information.
- Allow Fleet Controllers to monitor active buses.
- Identify buses experiencing significant delays or high crowding.
- Notify Fleet Controllers when configured delay or crowding thresholds are exceeded.

3. Stakeholders / Actors

The following actors are identified from the problem statement:

3.1 Commuter

The Commuter interacts with the system to obtain real-time information about buses. The commuter can view bus locations, estimated arrival times, and crowding levels for a selected route or bus.

3.2 Fleet Controller

The Fleet Controller is responsible for monitoring active buses and identifying operational issues such as significant delays and high passenger crowding. The controller can receive notifications about buses that exceed configured thresholds.

3.3 Bus GPS & Ticketing Sensor System

The Bus GPS & Ticketing Sensor System represents the external devices/systems that provide telemetry data to the platform. GPS sensors provide bus location information, while ticketing sensors provide passenger-related data used for crowding estimation.

4. Main System Functions

Based on the problem statement, the main functions of the system are:

1. Ingest bus GPS coordinates.
2. Track the current location of buses.
3. Calculate estimated arrival times for upcoming stops.
4. Receive ticketing sensor data.
5. Estimate passenger crowding levels.
6. Allow commuters to view bus information.
7. Allow Fleet Controllers to monitor active buses.
8. Detect significant delays or high crowding.
9. Notify Fleet Controllers about threshold violations.
10. Process telemetry from a large number of active buses simultaneously.

5. Key System Constraints

The system has the following important constraints:

- GPS coordinates must be processed every 5 seconds.
- The system must support telemetry from up to 1,000 active transit vehicles simultaneously.
- ETA calculations must use current bus and route information.
- Crowding estimates must be based on available ticketing sensor data.
- GPS and ticketing sensor data must be protected from unauthorized access.
- The system should maintain at least 99.5% availability during scheduled operating hours.

6. Functional Requirement Summary

The functional requirements describe what the system shall do.

The five functional requirements identified are:

- FR-001: Ingest GPS coordinates and recalculate ETAs.
- FR-002: Estimate passenger crowding using ticketing sensor data.
- FR-003: Allow commuters to view bus location, ETA, and crowding.
- FR-004: Allow Fleet Controllers to monitor active buses.
- FR-005: Notify Fleet Controllers about delay or crowding threshold violations.

7. Non-Functional Requirement Summary

The non-functional requirements describe how well the system should operate.

The two non-functional requirements identified are:

- NFR-001: Handle telemetry streams from up to 1,000 active vehicles simultaneously.
- NFR-002: Protect transit sensor data and maintain at least 99.5% system availability.

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall ingest bus GPS coordinates every 5 seconds and recalculate arrival ETAs for all upcoming route stops.	High	Pass: Valid GPS coordinates are received and the ETA for upcoming stops is recalculated within each 5second update cycle. Fail: GPS data is not processed or the ETA is not recalculated within the required update cycle.	Frequent GPS updates are essential for accurate live bus tracking and ETA calculation.	Given requirement from the problem statement.
FR-002	Functional	The system shall estimate the passenger crowding level of each active bus using data received from ticketing sensors.	High	Pass: The system calculates and displays the current crowding level when valid ticketing sensor data is available. Fail: The crowding level is not updated after valid sensor data is received.	Crowd information helps commuters choose suitable buses and helps controllers identify overcrowded services.	Ensure crowding levels are based on the latest available sensor data.

FR003	Functional	The system shall allow commuters to view the current location, estimated arrival time, and crowding level of buses operating on a selected route.	High	Pass: After selecting a route, the commuter can view the bus location, ETA, and crowding level. Fail: Any required information is missing despite valid data being available.	Providing live bus information is a primary function of the transit intelligence platform.	Displayed information should use the latest available data.
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Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-004	Functional	The system shall allow the Fleet Controller to monitor active buses and identify buses experiencing significant delays or high crowding levels.	Medium	Pass: The Fleet Controller can view active buses and identify buses exceeding configured delay or crowding thresholds. Fail: A bus exceeding a threshold is not identified.	Fleet monitoring enables timely identification of transport service problems.	Delay and crowding thresholds should be clearly defined.
FR-005	Functional	The system shall notify the Fleet Controller when an active bus exceeds a configured delay or crowding threshold.	Medium	Pass: A notification is generated when a bus exceeds the configured threshold. Fail: No notification is generated for a qualifying delay or crowding condition.	Automatic notifications help controllers respond quickly to significant service issues.	Notification should identify the affected bus and detected condition.

NFR001	Nonfunctional — Performance & Scalability	The tracking engine shall handle telemetry streams from up to 1,000 active transit vehicles simultaneously.	High	Pass: Load testing with 1,000 active vehicles confirms that telemetry is processed successfully without system failure. Fail: The system becomes unavailable or fails to process telemetry under the specified load.	The platform must support large-scale citywide bus operations during peak periods.	Given requirement from the problem statement.
NFR-002	Nonfunctional — Security & Availability	The system shall protect GPS and ticketing sensor data from	High	Pass: Security testing prevents unauthorized access to	Reliable and secure operation is necessary	Security and availability should be verified
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
		unauthorized access and maintain at least 99.5% availability during scheduled operating hours.		protected data and availability monitoring records at least 99.5% uptime. Fail: Unauthorized access occurs or availability falls below 99.5%.	because the system continuously processes transit telemetry and sensor data.	through testing and monitoring.

Requirement Validation

The requirements were reviewed against the following criteria:

Clarity

Each requirement uses the phrase "**The system shall...**" and describes a specific system behavior or constraint. **Testability**

Each requirement contains measurable **Pass/Fail acceptance criteria**, such as:

- 5-second GPS update interval.
- Support for 1,000 active vehicles.
- 99.5% availability.
- Successful notification when a threshold is exceeded.

Relevance

All requirements are directly derived from the given **Public Bus Live Tracking & Crowding Estimator** problem statement. **Priority**

High-priority requirements represent core system functionality or critical operational constraints. Medium-priority requirements support monitoring and operational response.

Conclusion

The problem statement describes a municipal transit intelligence platform that combines GPS telemetry and ticketing sensor data to provide real-time bus tracking, ETA estimation, and crowding information.

The requirements analysis identifies **five functional requirements and two non-functional requirements**. These requirements will be used to derive the system's actors and use cases for the UML use-case diagram and to prepare the use-case flow specification.

USE-CASE FLOW SPECIFICATION

Problem Statement #24 — Smart Cities, Transport & Logistics

Public Bus Live Tracking & Crowding Estimator

Use Case: UC-03 — View Bus Information

Primary Actor: Commuter

Goal:

Allow the commuter to view the current location, estimated arrival time (ETA), and crowding level of buses operating on a selected route.

Preconditions

1. The Public Bus Live Tracking & Crowding Estimator system is operational.
2. The commuter has selected a bus route or a specific bus.
3. Recent GPS and ticketing sensor data are available for the selected bus.

Postconditions

1. The commuter is shown the current location of the selected bus.
2. The commuter is shown the estimated arrival time at upcoming route stops.
3. The commuter is shown the estimated crowding level of the selected bus.

Main Success Scenario

1. The commuter selects a bus route or a specific bus.
2. The system retrieves the latest GPS coordinates of the selected bus.
3. The system determines the current location of the bus.
4. The system calculates the estimated arrival time (ETA) for upcoming route stops.
5. The system retrieves the latest passenger information from ticketing sensors.
6. The system estimates the current crowding level of the bus.
7. The system displays the bus location, ETA, and crowding level to the commuter.
8. The commuter views the updated bus information.
9. The use case ends successfully.

Alternate Flow

A1 — GPS Data Unavailable

1. The system detects that recent GPS data for the selected bus is unavailable.
2. The system cannot calculate a reliable current ETA.
3. The system displays a message informing the commuter that live ETA information is temporarily unavailable.
4. If previous bus information is available, the system displays the most recently known information.
5. The use case ends with limited information being displayed.

Use-Case Summary Use Case ID: UC-03

Use Case Name: View Bus Information

Primary Actor: Commuter

Related Requirements: FR-001, FR-002, FR-003

Purpose: To provide commuters with real-time bus location, estimated arrival time, and passenger crowding information.

Main Success Outcome: The commuter successfully views current bus location, ETA, and crowding information.

Alternate Outcome: If GPS data is unavailable, the system informs the commuter and provides the most recently available information where possible.

RTM (Requirements Traceability Matrix)

Requirement	Use Case	Actor	Test/Acceptance
FR-001	UC-01 Track Bus / ETA	GPS & Ticketing Sensor System	GPS received every 5 sec and ETA recalculated
FR-002	UC-02 Estimate Crowding	GPS & Ticketing Sensor System	Crowding updated from sensor data
FR-003	UC-03 View Bus Information	Commuter	Location, ETA and crowding displayed
FR-004	UC-04 Monitor Active Buses	Fleet Controller	Active buses and threshold violations visible
FR-005	UC-05 Receive Alerts	Fleet Controller	Notification generated when threshold exceeded
NFR-001	System-wide	System	Supports 1,000 active vehicles
NFR-002	System-wide	System/Admin	Security maintained and $\geq 99.5\%$ availability